

Task 1:

Q1

```
def slargest(l):  
    nums=set(l)  
    if len(nums)<2:  
        return -1  
    m=max(nums)  
    nums.remove(m)  
    return max(nums)  
  
l=[12,35,1,10,34,1]  
print(slargest(l))
```

Q2

```
def rarr(nums, k):  
    n = len(nums)  
    k %= n  
    temp = nums[-k:] + nums[:-k]  
    for i in range(n):  
        nums[i] = temp[i]  
  
l = [1,2,3,4,5,6,7]  
k = 3  
  
rarr(l,k)  
print(l)
```

q3

```
bool Anagram(string s, string t) {
    vector<int> freq(26, 0);

    for (char c : s) {
        if (c != ' ') {
            c = tolower(c);
            freq[c - 'a']++;
        }
    }
    for (char c : t) {
        if (c != ' ') {
            c = tolower(c);
            freq[c - 'a']--;
        }
    }
    for (int x : freq) {
        if (x != 0)
            return false;
    }
    return true;
}
```

Q4

```
int lengthOfLongestSubstring(string s) {
    unordered_map<char, int> freq;
    int l = 0, maxLen = 0;

    for (int r=0;r<s.size();r++) {
        char c=s[r];
        freq[c]++;
        while (freq[c]>1) {
            freq[s[l]]--;
            l++;
        }

        maxLen = max(maxLen,r-l+1);
    }
    return maxLen;
}
```

Q5

```
SELECT d.dept_name, e.name, e.salary FROM Employees e JOIN Departments d ON  
e.dept_id = d.dept_id  
WHERE (e.dept_id, e.salary) IN ( SELECT dept_id, MAX(salary) FROM Employees  
GROUP BY dept_id);
```

Q6

```
SELECT DATE_FORMAT(order_date, '%b') AS month, SUM(amount) AS  
total_revenue, COUNT(*) AS total_orders  
FROM Orders WHERE YEAR(order_date) = 2024 GROUP BY DATE_FORMAT(order_date,  
'%b') ORDER BY total_revenue DESC;
```